THREE BASIC PROPERTIES OF A NOETHERIAN TOPOLOGICAL SPACE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that every irreducible closed subset of a Noetherian space is a finite union of its irreducible components. Invoked inside the counted proof (source lines 1505-1508). Closes the minimal-counterexample argument for the finiteness of the components.
- The statement that an irredundant finite union of closed subsets has exactly those subsets as its irreducible components. Invoked near the end of the counted proof (source lines 1510-1512). Turns the finite union into the decomposition into irreducible components.

Source reference. Original lemma lemma-irreducible: sources/stacks-topology/pinned-original-source.tex, source lines 1103–1103. Navigation pointer only: the counted text refers to the Stacks item lemma-irreducible, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-pick-irreducible-components: sources/stacks-topology/pinned-original-source.tex, source lines 1142–1142. Navigation pointer only: the counted text refers to the Stacks item lemma-pick-irreducible-components, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a Noetherian topological space.*

(1) *Any subset of X with the induced topology is Noetherian.*
(2) *The space X has finitely many irreducible components.*

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(3) *Each irreducible component of X contains a nonempty open of X.*

Proof. Let $T \subset X$ be a subset of X. Let $T_1 \supset T_2 \supset \ldots$ be a descending chain of closed subsets of T. Write $T_i = T \cap Z_i$ with $Z_i \subset X$ closed. Consider the descending chain of closed subsets $Z_1 \supset Z_1 \cap Z_2 \supset Z_1 \cap Z_2 \cap Z_3 \ldots$ This stabilizes by assumption and hence the original sequence of T_i stabilizes. Thus T is Noetherian.

Let A be the set of closed subsets of X which do not have finitely many irreducible components. Assume that A is not empty to arrive at a contradiction. The set A is partially ordered by inclusion: $\alpha \leq \alpha' \Leftrightarrow Z_\alpha \subset Z_{\alpha'}$. By the descending chain condition we may find a smallest element of A, say Z. As Z is not a finite union of irreducible components, it is not irreducible. Hence we can write $Z = Z' \cup Z''$ and both are strictly smaller closed subsets. By construction $Z' = \bigcup Z'_i$ and $Z'' = \bigcup Z''_j$ are finite unions of their irreducible components (Lemma lemma-irreducible). Hence $Z = \bigcup Z'_i \cup \bigcup Z''_j$ is a finite union of irreducible closed subsets. After removing redundant members of this expression, this will be the decomposition of Z into its irreducible components (Lemma lemma-pick-irreducible-components), a contradiction.

Let $Z \subset X$ be an irreducible component of X. Let $Z_1, \ldots, Z_n$ be the other irreducible components of X. Consider $U = Z \setminus (Z_1 \cup \ldots \cup Z_n)$. This is not empty since otherwise the irreducible space Z would be contained in one of the other Z_i. Because $X = Z \cup Z_1 \cup \ldots Z_n$ (see Lemma lemma-irreducible), also $U = X \setminus (Z_1 \cup \ldots \cup Z_n)$ and hence open in X. Thus Z contains a nonempty open of X. $\square$